

**INTERGRATED COMPLIANCE TRACKING AND RISK
MANAGEMENT SYSTEM FOR FACILITY SAFETY**

By

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**A Dissertation Submitted in fulfillment of the requirements for the degree of Bachelor
of Science Information system management of Ardhi University**

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DECLARATION

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The undersigned certify that they have read and hereby recommend for acceptance by Ardhi University a dissertation titled: integrated compliance tracking and risk management system for facility safety, in fulfillment of the requirements for the degree of Bachelor of Science Information system management of Ardhi University.

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LIST OF ABBREVIATIONS

ARU	Ardhi University
OS	Operating Systems
UML	Unified Modeling Language
ERD	Entity Relationship Diagram
HESLB	Higher Education Students' Loans Board

ABSTRACT

This project presents the design and development of an *Integrated Compliance Tracking and Risk Management System for Facility Safety*. The system aims to address challenges associated with manual and fragmented approaches to monitoring facility compliance, inspection schedules, and risk management. Many organizations lack centralized systems to track statutory inspections, certification expiry, and asset-based risks, which may lead to non-compliance, safety incidents, and inefficient operations. The proposed system provides a digital solution to enhance proactive monitoring and decision-making in facility safety management.

The system was developed using a web-based architecture that integrates key modules such as asset management, inspection scheduling, compliance tracking, risk assessment, incident reporting, and corrective and preventive action (CAPA) management. Data was collected through literature review and analysis of existing facility management practices. Modern technologies including React, Node.js, and PostgreSQL were used to design and implement the system. The system automates inspection reminders, tracks compliance status, maintains risk registers, and generates alerts for expiring certifications and unresolved issues.

The outcome of this project is a centralized platform that improves visibility, accountability, and efficiency in managing facility safety and regulatory compliance. The system supports timely decision-making, reduces the risk of non-compliance, and enhances overall safety performance. It is expected to benefit organizations by providing a structured and reliable approach to facility compliance and risk management.

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CHAPTER ONE

INTRODUCTION

1.1 General Introduction

Safety and facilities compliance management is a significant aspect of organizations dealing with complex facilities such as fire protection systems, lifts, boilers, pressure vessels, and electrical equipment. Therefore, proper monitoring of such facilities is a prerequisite for the prevention of accidents and the maintenance of continuity in the facilities. However, traditional safety management is usually manual, disjointed, and paper-intensive, which may result in delayed reporting of hazards, missed inspections, and ineffective corrective actions. In order to improve the safety and facilities compliance management of an organization, a centralized digital facility safety management system is required. Such a facility safety management system would ensure the integration of compliance tracking, risk management, and corrective actions in a single facility safety management system. Digital monitoring tools, asset-based risk registers, and automated alerts are some of the modern technologies required for the effective implementation of a facility safety management system. (Marchetti, 2017).

The proposed Integrated Compliance Tracking and Risk Management System for Facility Safety provides a unified solution to manage safety data, track asset risks, schedule inspections, and monitor compliance efficiently.

1.2 Statement of the Problem

In many organizations, facility safety and compliance management are still handled using manual, paper-based, or fragmented systems. These approaches make it difficult to maintain accurate records, track inspections, and manage safety-related activities efficiently. As a result, organizations often struggle to ensure that all facility assets such as fire systems, lifts, electrical installations, boilers, and pressure vessels meet required safety and regulatory standards.

However, the absence of a centralized and integrated system creates significant challenges in managing facility operations. Organizations face poor record keeping, missed statutory inspections, delayed hazard and incident reporting, and limited visibility of compliance status. In addition, there is no effective mechanism for real-time risk identification, monitoring of asset-based risks, or tracking of corrective and preventive actions, which increases the likelihood of safety incidents and regulatory non-compliance.

Therefore, this study proposes the development of an Integrated Compliance Tracking and Risk Management System for Facility Safety. The system will provide a centralized platform for

managing compliance information, tracking inspections and certifications, supporting asset-based risk assessment, and monitoring corrective actions. This will improve efficiency, enhance safety management, and ensure regulatory compliance within facilities.

1.3 Objectives

1.3.1 General Objective

To develop an Integrated Compliance Tracking and Risk Management System for Facility Safety.

1.3.2 Specific objective

- i. To identify and gather user requirements for the Integrated Compliance Tracking and Risk Management System for Facility Safety.
- ii. To design the Integrated Compliance Tracking and Risk Management System for Facility Safety.
- iii. To implement the Integrated Compliance Tracking and Risk Management System for Facility Safety.
- iv. To test and validate the Integrated Compliance Tracking and Risk Management System for Facility Safety.

1.4 Research Questions

Following the identification of the problem with digital and facility level compliance monitoring, the research questions for the study include:

- i. How can a digital solution be developed to support statutory inspections and compliance management for the assets at a facility, such as fire safety, lifts, electrical systems, boilers, and pressure vessels?
- ii. How can a risk register be developed with a link between the assets at a facility and the associated hazard ratings and mitigation strategies?
- iii. How can the digital solution for scheduling inspections and the expiration of certifications help with regulatory compliance and prevent any possible non-compliance?
- iv. How can the real-time reporting of hazards and the digital CAPA processes help with the proactive management of safety at a facility?

1.5 Significance of the Study

The Integrated Compliance Tracking and Risk Management System for Facility Safety is significant because it addresses critical gaps in current facility management practices, where

safety compliance and risk tracking are often manual, fragmented, or paper-based. By centralizing facility data and automating monitoring processes, the system provides several key benefits:

- i. **Enhanced Safety and Risk Management:** The system enables real time hazard monitoring, asset-level risk tracking, and timely corrective and preventive actions (CAPA), reducing the accidents and improving overall safety within facilities.
- ii. **Regulatory Compliance:** By tracking statutory inspections, certification expiries, the system ensures adherence to local and international safety standards, supporting organizations in avoiding penalties and legal issues.
- iii. **Operational Efficiency:** Automating compliance monitoring and record keeping reduces manual workload, prevents delays in hazard reporting, and improves decision-making efficiency for facility managers.
- iv. **Accountability and Transparency:** The platform provides a centralized repository of safety records, risk assessments, and corrective actions, enhancing accountability across departments and facilitating audit readiness.

1.6 Structure of the report

This report is organized into seven chapters, each addressing a specific aspect of the study. Chapter One presents the introduction, which includes the background of the study, problem statement, research objectives, and scope of the project. It provides the overall context and purpose of the study.

Chapter Two contains the literature review, which examines existing studies, technologies, and approaches related to facility compliance tracking and risk management. This chapter helps identify research gaps and justifies the need for the proposed system.

Chapter Three describes the research methodology used in this study. It explains the data collection methods, system development approach, and tools and techniques applied during the development process.

Chapter Four focuses on system analysis and design. It presents system requirements, system architecture, and the design of key components such as modules, data structures, and user interfaces.

Chapter Five discusses system implementation and testing. It explains how the system was developed, integrated, and tested to ensure it meets the required functionality and performance standards.

Chapter Six presents the results and discussion. It evaluates the performance of the system, analyzes outcomes, and discusses how the system addresses the identified problem.

Finally, Chapter Seven provides the conclusion and recommendations. It summarizes the findings of the study and suggests possible improvements and future work.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

A literature review is a structured analysis of existing research and scholarly work related to a particular field of study. It helps to identify previous efforts, existing challenges, and knowledge gaps while providing a strong foundation for new projects (Sommerville, 2011). For this study, the literature review focuses on facilities compliance monitoring, facility risk tracking, and facilities records keeping.

2.2 Related Studies

Abad et al. (2022) examined the application of digital technologies to improve building and facility safety. The study highlighted the use of Building Information Modelling (BIM), digital monitoring systems, and smart technologies to enhance safety visibility and real-time hazard identification. Findings showed that integrating digital models with real-time data improves risk awareness and supports informed decision-making in facility management. However, the study has limitations. It does not provide a structured framework for statutory compliance monitoring, inspection scheduling, certification tracking, or Corrective and Preventive Action (CAPA) workflows. Additionally, asset-based risk registers and automated compliance alerts were not addressed, revealing a gap in centralized compliance tracking and regulatory management.

Marchetti (2017) examined integrated risk and safety management systems within engineering and industrial environments. The chapter emphasized unifying risk assessment, monitoring, and control processes into a comprehensive framework to enhance organizational safety. It highlighted structured risk assessment models, digital information systems, and safety documentation platforms to improve hazard identification, coordination, and decision making. The findings suggested that centralized integration strengthens transparency and risk mitigation effectiveness. However, the chapter is largely conceptual and lacks implementation detail. It does not provide mechanisms for real time hazard monitoring, asset-based inspection tracking, compliance scheduling, or automated corrective action workflows, revealing a gap between theoretical integration and practical facility level safety management.

Ramos et al. (2020) examined environmental and safety compliance management systems within organizational settings, focusing on structured frameworks, internal auditing, and governance mechanisms to support regulatory adherence. The study highlights that

organizations applying systematic compliance monitoring demonstrate improved regulatory performance, reduced environmental violations, and enhanced accountability in safety management processes. However, limitations exist in facility level and digital compliance management. The research primarily addresses policy and administrative compliance rather than operational monitoring of assets such as fire protections, lifts, electrical installations, boilers, and pressure vessels. Additionally, it lacks real time hazard detection, automated compliance alerts, inspection scheduling models, and integrated asset-based risk registers. While corrective actions are conceptually discussed, no structured digital workflows for task assignment or deadline monitoring are provided, indicating a gap in proactive facility safety management.

Chu et al. (2018) examined energy and water utility infrastructure management (WIM) systems, focusing on integrated monitoring technologies and control platforms to improve operational efficiency and reliability of facility utility services. The study highlights that such monitoring enhances system visibility, supports timely maintenance decisions, and contributes to safer and more effective facility operations. However, limitations exist in compliance tracking and structured risk management. The research primarily addresses operational monitoring rather than statutory inspection compliance, certification expiry management, or regulatory governance. Additionally, it lacks an integrated asset-based risk register linking facility components to hazard ratings, as well as automated Corrective and Preventive Action (CAPA) workflows for addressing safety issues. While operational insights are provided, no centralized compliance tracking system or structured digital workflow for continuous facility safety assurance is established, indicating a gap in proactive regulatory and safety management.

Abad et al. (2020) examined an Integrated Management System (IMS) to improve OH&S risk management in a medium-sized waste firm. The study highlights that digital documentation and structured risk assessment reduced occupational accidents, increased worker engagement, and strengthened management support. However, limitations exist at the facility level, as the research does not cover assets like fire protections, electrical systems, boilers, or pressure vessels. It also lacks real-time compliance tracking, digital data consolidation, and automated CAPA workflows, indicating a gap between IMS theory and practical facility level hazard monitoring.

Salaheldin et al. (2021) examined a fire code compliance assessment tool for educational facilities, developing a structured worksheet based on international fire codes to assess fire prevention measures. The study highlights that systematic evaluation of 34 prevention

measures helps identify shortcomings such as inactive record keeping and design non conformities, supporting thorough inspections. However, limitations include lack of digital automation, real-time hazard detection, compliance alerting, risk scoring algorithms, and integration with digital asset monitoring. The research focuses on paper-based checklist methodology rather than technologies such as mobile inspection apps, sensor data, or automated dashboards, indicating a gap in proactive digital fire compliance workflows.

Negi et al. (2024) reviewed the integration of *Industry 4.0 technologies* (such as IoT sensors, cloud computing, big data analytics, artificial intelligence/machine learning, edge computing, blockchain, and Building Information Modelling) in fire and safety management to enhance hazard detection, monitoring, and response. The study highlights that these technologies can enable real time data collection, predictive analytics, and situational awareness for fire risk reduction. However, limitations exist in proposing concrete frameworks for automated compliance alerts, structured asset based risk registers, inspection scheduling models, and CAPA workflows, indicating that operational governance and regulatory tracking remain less developed than the core technological potential.

Amer et al. (2022) examined electrical fire risks in substations and proposed a risk management framework emphasizing hazard identification, causes of electrical fires (e.g., insulation failure, arc flash, equipment overheating), and mitigation strategies, often referencing NFPA and IEEE standards. The study highlights that these risk evaluation practices support safety planning and decision making at facilities with high voltage equipment. However, limitations include absence of digital compliance tracking technologies such as sensor-based condition monitoring, automated alert systems, structured inspection scheduling tools, or integrated digital risk registers, indicating that the proposed framework remains largely conceptual and manual in execution.

Shabani et al. (2024) provided a comprehensive review of safety audits and their role in ensuring workplace safety and compliance in Zimbabwe, synthesizing evidence from multiple databases on audit processes, components, benefits, and limitations. The study highlights that safety audits systematically evaluate organizational safety management systems, policies, and practices to identify hazards and enable corrective actions that prevent accidents and occupational illnesses. However, limitations exist in digital automation and advanced technologies; the review focuses on traditional audit frameworks rather than technologies such as real time monitoring tools, sensor-based hazard detection, automated compliance alerts, or

integrated digital risk registers, indicating a gap in modern, technology enabled compliance systems.

2.3 Research Gap

Despite advances in digital monitoring, integrated management systems, and compliance frameworks, several gaps remain in facility level safety and compliance management, as highlighted in the literature. This project aims to bridge these gaps by designing an Integrated Compliance Tracking and Risk Management System for Facility Safety that provides the following improvements:

- i. **Lack of real time hazard monitoring:** Existing frameworks are mostly not focused on real time hazard monitoring. The proposed system enables to identified risks before happen by alert the person or team responsible for maintenances.
- ii. **Absence of asset-based risk tracking:** Current IMS or compliance systems do not track individual facility assets such as fire systems, lifts, boilers, or electrical installations. Our system creates dynamic risk registers for each asset, updating hazard levels continuously.
- iii. **No automated corrective and preventive action (CAPA) workflows:** Literature shows CAPA processes are mostly manual or undefined. This system assigns, tracks, and escalates corrective actions automatically with deadlines and responsible persons.
- iv. **Limited statutory compliance monitoring:** Existing studies do not integrate inspection scheduling, certification expiry alerts, or regulatory reporting. Our system centralizes all

2.3 Summary of literature reviews

S/N	Author's name& year	Technologies used					
		Compliance Framework	Manual Checklist	IoT / Smart Sensors	Digital Monitoring	Risk Register	Automated Alerts/ CAPA
1	(Ramos et al., 2020)	✗	✗	✓	✓	✗	✗
2	(Chu et al., 2018)	✗	✗	✓	✓	✗	✗
3	(Abad et al., 2020)	✓	✗	✗	✓	✗	✗
4	(Salaheldin et al., 2021)	✓	✓	✗	✗	✗	✗
5	(Negi et al., 2024)	✗	✗	✓	✓	✗	✗
6	(Amer et al., 2022)	✓	✓	✗	✗	✗	✗
7	(Shabani et al., 2024)	✓	✓	✗	✗	✗	✗
8	(IEOM Study, 2022)	✓	✗	✗	✗	✓	✗
1	(Tarima & Afidhu,2026)	✓	✗	✗	✓	✓	✓

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter presents the methodology used in the development of the Integrated Compliance Tracking and Risk Management System for Facility Safety. It describes the software development approach, data collection methods, system design techniques, implementation technologies, and testing procedures. The study follows the Software Development Life Cycle (SDLC) to ensure a structured and systematic development process (Pressman, 2014; Sommerville, 2011).

3.2 Selected Methodology

The Agile methodology was adopted as the general approach for the development of the Integrated Compliance Tracking and Risk Management System for Facility Safety. Agile is an iterative and incremental software development approach that emphasizes flexibility, continuous improvement, stakeholder collaboration, and regular feedback. Instead of following a rigid linear process, the project was divided into small development cycles known as sprints, where each sprint delivered a functional component of the system.

Agile was selected because it allows adaptability throughout the development process, making it easier to incorporate feedback from facility managers, safety officers, and compliance stakeholders. This is particularly important for a compliance and risk management system, where requirements may evolve as regulations and operational needs change. Agile also supports continuous testing, early detection of issues, and incremental delivery of system features, ensuring a practical, user centered, and efficient solution.

4.2.1 Methodology for Gathering User Requirement

To accurately capture the needs of end users and stakeholders, the following methods were used:

Literature Review: This involved examining previous studies, journals, and existing systems related to facility compliance, safety management, and risk assessment. It provided insight into current challenges, existing solutions, and best practices in compliance tracking and risk management systems. Literature review provided foundational knowledge and helped in identifying research gaps, ensuring that the proposed system aligns with modern approaches and industry requirements.

Interviews: Structured interviews were conducted with facility managers, safety officers, and technical personnel to understand their requirements, challenges, and expectations regarding compliance tracking and risk management. Interviews enabled the collection of detailed and practical user requirements based on real world experiences, improving the relevance and usability of the system.

Observation: Observations were carried out within facility environments to study existing manual or semi digital processes used for inspections, compliance tracking, and risk management. Observation helped identify inefficiencies, delays, and gaps in current systems, providing practical insights for designing an improved and more effective digital solution.

4.2.2 Methodology for System design

For system design, modeling and visualization techniques such as Entity-Relationship Diagrams (ERDs), Unified Modeling Language (UML), and User Interface (UI) mockups were used to clearly represent the structure and functionality of the system.

ERDs (Entity-Relationship Diagrams): These diagrams are used to represent the structure of the database by showing entities such as users, assets, inspections, risks, and their relationships. ERDs help ensure a well-structured and normalized database design, which improves data organization and system efficiency.

UML (Unified Modeling Language): UML is a standardized modeling language used to visualize system design through diagrams such as Use Case, Class, Sequence, and Activity diagrams. UML helps in understanding system behavior, user interactions, and data flow, making it easier to design and communicate system functionality.

Mockups (UI/UX Design Tools): Tools such as Draw.io and Figma were used to design and visualize the user interface of the system before implementation. Mockups help in planning the system layout and user experience, ensuring that the system is user friendly and easy to navigate.

4.2.3 Methodology for System implementation

For system implementation, modern web technologies and development tools were used to ensure the system is efficient, scalable, and user friendly.

React (Web Framework): Used to develop the system interface for users such as administrators, inspectors, and technicians. Because React supports the creation of interactive and responsive user interfaces, improving usability and system performance.

JavaScript/TypeScript: Used for implementing system logic and handling interactions between different components. Since These languages provide flexibility and support modern development practices, ensuring efficient and scalable system functionality.

Node.js and Express: Used for backend development and creation of APIs that connect the frontend to the database. Since They enable fast processing of requests and simplify the development of scalable and efficient server-side applications.

PostgreSQL: Used as the database for storing system data such as assets, inspections, risks, and compliance records. Since it ensures reliable, secure, and structured storage of relational data required for system operations.

Postman: Used for testing system APIs and verifying communication between system components. Since helps detect errors early and ensures that all system functionalities work correctly.

Git and GitHub: Used for version control and project management during development. Because they help track changes, manage code versions, and support collaborative development.

4.2.4 Methodology for System testing and Validation

To ensure the system functions correctly and meets user requirements, the following methods were used for system testing and validation:

Unit Testing: This involves testing individual components or modules of the system such as asset management, inspection scheduling, and risk assessment independently. Unit testing helps identify and fix errors at an early stage, ensuring that each module performs its intended function correctly.

Integration Testing: This involves testing how different system components interact with each other, such as the connection between the frontend, backend, and database. Integration testing ensures that all modules work together seamlessly and that data flows correctly across the system.

3.1 Methodology table

Table 3.1 Methodology Table

S/ N	specific objective	methodology	tools	deliverable
1	To analyse User requirements for the Integrated Compliance Tracking and Risk Management System for Facility Safety	Questionnaire	Paper based form and online sources (Google forms)	User requirement specification document
2	To design the Integrated Compliance Tracking and Risk Management System for Facility Safety	Structured design methodology	Draw io, balsamiq and Pencil	System design document (SDD)
3	To develop the Integrated Compliance Tracking and Risk Management System for Facility Safety	Agile methodology	Visual studio code, React.js	Function software application (FSA)
4	To evaluate the Integrated Compliance Tracking and Risk Management System for Facility Safety	Alpha testing	Selenium	Integrated Compliance Tracking and Risk Management System for facility

CHAPTER FOUR

SYSTEM ANALYSIS AND DESIGN

4.1 Introduction

This chapter presents the system analysis and design of the Integrated Compliance Tracking and Risk Management System for Facility Safety. It describes the system requirements, design methodologies, and overall system structure. The chapter also explains how different system components interact to support compliance tracking, risk assessment, and inspection management. It provides a foundation for understanding how the system will be developed and implemented to meet user needs.

4.2 Requirement Analysis

Requirement analysis is the process of identifying, documenting, and validating the needs and expectations of users and stakeholders for a system (Somerville, 2011). It ensures that the developed system fulfills its intended purpose and operates effectively within its environment. In this study, requirement analysis is categorized into two main types: system requirements and user requirements. System requirements define the technical and operational resources needed for the system to function, while user requirements describe how users interact with the system and the services it must provide.

4.2.1 System Requirement

System requirements refer to the specifications that define the technical infrastructure and resources necessary for the operation of a software system (IEEE, 1998). They describe what is needed for the system to function effectively within its environment and ensure reliable performance. In this study, system requirements are categorized into three main types: hardware requirements, software requirements, and data requirements, each of which plays a critical role in supporting the development and operation of the system.

4.2.1.1 Data Requirement

Data requirements describe the specific data elements that the system needs to collect, process, store, and retrieve to support its operations. In this study, data was collected through literature review, interviews with facility personnel, and observation of existing compliance and safety management practices. These sources provided both theoretical and practical insights into the types of data required for effective facility compliance tracking and risk management.

The types of data used in the system include

- i. **User Data:** Includes information such as name, role (administrator, inspector, technician), contact details, and login credentials. This data is used for user authentication, authorization, and tracking system activities.
- ii. **Asset Data:** Includes details of facility assets such as fire protection systems, lifts, electrical installations, boilers, and pressure vessels. This data is used to monitor asset status and ensure compliance with safety regulations.
- iii. **Inspection Data:** Includes inspection schedules, dates, inspection results, and status (completed, pending, overdue). This data is used to track inspections and ensure they are conducted within the required timeframes.
- iv. **Compliance Data:** Includes certification records, expiry dates, and compliance status of assets. This data is used to monitor regulatory compliance and generate alerts for expiring or overdue certifications.
- v. **Risk Data:** Includes identified hazards, risk levels, and mitigation measures linked to specific assets. This data is used for risk assessment and management.
- vi. **Incident Data:** Includes records of incidents and near misses, including their causes, impact, and affected assets. This data is used to analyze safety issues and prevent future occurrences.
- vii. **Task (CAPA) Data:** Includes corrective and preventive actions, assigned personnel, deadlines, and task status. This data is used to track the resolution of identified issues and ensure accountability.

4.2.1.2 Software Requirement

Software requirements refer to the specific programs, frameworks, and tools needed to develop, run, and maintain a software system effectively (Sommerville, 2011). They define the technological environment within which the system operates and ensure that all components function correctly and efficiently.

The software used in this project includes:

- i. **React:** A JavaScript library used to develop the web-based user interface. It enables the creation of interactive and responsive components for users such as administrators, inspectors, and technicians.

- ii. **JavaScript/TypeScript:** Programming languages used to implement application logic, handle user interactions, and manage system functionality on both the frontend and backend.
- iii. **Node.js and Express:** Used for backend development and creating APIs that handle communication between the user interface and the database. They support efficient data processing and system operations.
- iv. **PostgreSQL:** A relational database management system used to store and manage system data such as assets, inspections, risks, and compliance records in a structured format.
- v. **Visual Studio Code:** A code editor used for writing, editing, and debugging system code during development. It provides useful extensions and tools that improve productivity.
- vi. **Postman:** A tool used for testing APIs to ensure proper communication between the frontend and backend components of the system.
- vii. **Git and GitHub:** Used for version control and managing the development process. They help track code changes, support collaboration, and maintain backups of the project.

4.2.1.3 Hardware Requirement

Hardware requirements refer to the physical devices and equipment needed to support the development, deployment, and operation of a software system (Shelly & Rosenblatt, 2012). These requirements ensure that the system can run efficiently and that users are able to access and interact with it effectively.

The hardware used in this project includes:

- i. **Laptop/PC:** Used for system development, coding, and testing. It provides the computing power required to run development tools, servers, and databases.
- ii. **Server/Cloud Infrastructure:** Used to host the application and database, enabling the system to be accessed remotely by users. It ensures data availability and system scalability.
- iii. **Mobile Devices (Smartphones/Tablets):** Used by inspectors and technicians to access the system in real-time, especially during inspections and reporting activities in the field.
- iv. **Internet Connection:** Required for communication between the client (user interface) and the server, as well as for accessing cloud-based services and databases.

- v. These hardware components collectively support the effective development, deployment,

4.2.2 User Requirement

User requirements describe the needs and expectations of users regarding what the system should do in order to achieve its intended purpose. They focus on the services and interactions required by end users and stakeholders. According to Sommerville (2011), user requirements are typically divided into two categories: functional requirements and nonfunctional requirements, which define what the system does and how well it performs those functions.

4.2.2.1 Functional Requirement

Functional requirements describe the specific services and behaviors that the system must provide to users. According to Pressman (2014), they define the operations that the system should perform to meet user needs and solve the identified problem.

The Facility Compliance and Safety Management System shall:

- i. Allow users (administrators, inspectors, and technicians) to register, log in, and manage their accounts securely.
- ii. Enable administrators to register and manage facility assets such as fire systems, lifts, electrical installations, boilers, and pressure vessels.
- iii. Allow the system to schedule inspections and track inspection status (completed, pending, or overdue).
- iv. Enable inspectors to record inspection results and report non-compliance issues.
- v. Generate alerts and notifications for upcoming inspections, overdue tasks, and expiring certifications.
- vi. Provide a risk assessment module for identifying hazards, assessing risks, and managing risk registers linked to assets.

4.2.2.2 Nonfunctional Requirement

Nonfunctional requirements describe the quality attributes of the system, including performance, usability, security, and reliability. According to Sommerville (2011), they define how well the system performs its functions rather than what it does.

The system must:

- i. Ensure data security and privacy through secure authentication and role-based access control.
- ii. Provide high performance with fast response time during data processing and retrieval.
- iii. Be user-friendly with a simple, clear, and responsive interface.
- iv. Ensure system reliability and availability for continuous monitoring of compliance activities.
- v. Be scalable to support additional users, assets, and features as the system grows.

CHAPTER FOUR

SYSTEM ANALYSIS AND DESIGN

1.7 Introduction

This chapter presents the system analysis and design of the Integrated Compliance Tracking and Risk Management System for Facility Safety. It describes the system requirements, design methodologies, and overall system structure. The chapter also explains how different system components interact to support compliance tracking, risk assessment, and inspection management. It provides a foundation for understanding how the system will be developed and implemented to meet user needs.

1.8 Requirement Analysis

Requirement analysis is the process of identifying, documenting, and validating the needs and expectations of users and stakeholders for a system (Somerville, 2011). It ensures that the developed system fulfills its intended purpose and operates effectively within its environment. In this study, requirement analysis is categorized into two main types: system requirements and user requirements. System requirements define the technical and operational resources needed for the system to function, while user requirements describe how users interact with the system and the services it must provide.

2 System Requirement

System requirements refer to the specifications that define the technical infrastructure and resources necessary for the operation of a software system (IEEE, 1998). They describe what is needed for the system to function effectively within its environment and ensure reliable performance. In this study, system requirements are categorized into three main types: hardware requirements, software requirements, and data requirements, each of which plays a critical role in supporting the development and operation of the system.

1 Data Requirement

Data requirements describe the specific data elements that the system needs to collect, process, store, and retrieve to support its operations. In this study, data was collected through literature review, interviews with facility personnel, and observation of existing compliance and safety management practices. These sources provided both theoretical and practical insights into the types of data required for effective facility compliance tracking and risk management.

The types of data used in the system include

- 2.1 **User Data:** Includes information such as name, role (administrator, inspector, technician), contact details, and login credentials. This data is used for user authentication, authorization, and tracking system activities.
- 2.2 **Asset Data:** Includes details of facility assets such as fire protection systems, lifts, electrical installations, boilers, and pressure vessels. This data is used to monitor asset status and ensure compliance with safety regulations.
- 2.3 **Inspection Data:** Includes inspection schedules, dates, inspection results, and status (completed, pending, overdue). This data is used to track inspections and ensure they are conducted within the required timeframes.
- 2.4 **Compliance Data:** Includes certification records, expiry dates, and compliance status of assets. This data is used to monitor regulatory compliance and generate alerts for expiring or overdue certifications.
- 2.5 **Risk Data:** Includes identified hazards, risk levels, and mitigation measures linked to specific assets. This data is used for risk assessment and management.
- 2.6 **Incident Data:** Includes records of incidents and near misses, including their causes, impact, and affected assets. This data is used to analyze safety issues and prevent future occurrences.
- 2.7 **Task (CAPA) Data:** Includes corrective and preventive actions, assigned personnel, deadlines, and task status. This data is used to track the resolution of identified issues and ensure accountability.

2 Software Requirement

Software requirements refer to the specific programs, frameworks, and tools needed to develop, run, and maintain a software system effectively (Sommerville, 2011). They define the technological environment within which the system operates and ensure that all components function correctly and efficiently.

The software used in this project includes:

- 2 **React:** A JavaScript library used to develop the web-based user interface. It enables the creation of interactive and responsive components for users such as administrators, inspectors, and technicians.
- 3 **JavaScript/TypeScript:** Programming languages used to implement application logic, handle user interactions, and manage system functionality on both the frontend and backend.
- 4 **Node.js and Express:** Used for backend development and creating APIs that handle communication between the user interface and the database. They support efficient data processing and system operations.
- 5 **PostgreSQL:** A relational database management system used to store and manage system data such as assets, inspections, risks, and compliance records in a structured format.
- 6 **Visual Studio Code:** A code editor used for writing, editing, and debugging system code during development. It provides useful extensions and tools that improve productivity.
- 7 **Postman:** A tool used for testing APIs to ensure proper communication between the frontend and backend components of the system.
- 8 **Git and GitHub:** Used for version control and managing the development process. They help track code changes, support collaboration, and maintain backups of the project.

3 Hardware Requirement

Hardware requirements refer to the physical devices and equipment needed to support the development, deployment, and operation of a software system (Shelly & Rosenblatt, 2012). These requirements ensure that the system can run efficiently and that users are able to access and interact with it effectively.

The hardware used in this project includes:

- 1 **Laptop/PC:** Used for system development, coding, and testing. It provides the computing power required to run development tools, servers, and databases.
- 2 **Server/Cloud Infrastructure:** Used to host the application and database, enabling the system to be accessed remotely by users. It ensures data availability and system scalability.
- 3 **Mobile Devices (Smartphones/Tablets):** Used by inspectors and technicians to access the system in real-time, especially during inspections and reporting activities in the field.
- 4 **Internet Connection:** Required for communication between the client (user interface) and the server, as well as for accessing cloud-based services and databases.

- 5 These hardware components collectively support the effective development, deployment,

3 User Requirement

User requirements describe the needs and expectations of users regarding what the system should do in order to achieve its intended purpose. They focus on the services and interactions required by end users and stakeholders. According to Sommerville (2011), user requirements are typically divided into two categories: functional requirements and nonfunctional requirements, which define what the system does and how well it performs those functions.

2 Functional Requirement

Functional requirements describe the specific services and behaviors that the system must provide to users. According to Pressman (2014), they define the operations that the system should perform to meet user needs and solve the identified problem.

The Facility Compliance and Safety Management System shall:

- 1 Allow users (administrators, inspectors, and technicians) to register, log in, and manage their accounts securely.
- 2 Enable administrators to register and manage facility assets such as fire systems, lifts, electrical installations, boilers, and pressure vessels.
- 3 Allow the system to schedule inspections and track inspection status (completed, pending, or overdue).
- 4 Enable inspectors to record inspection results and report non-compliance issues.
- 5 Generate alerts and notifications for upcoming inspections, overdue tasks, and expiring certifications.
- 6 Provide a risk assessment module for identifying hazards, assessing risks, and managing risk registers linked to assets.

3 Nonfunctional Requirement

Nonfunctional requirements describe the quality attributes of the system, including performance, usability, security, and reliability. According to Sommerville (2011), they define how well the system performs its functions rather than what it does.

The system must:

- 1 Ensure data security and privacy through secure authentication and role-based access control.
- 2 Provide high performance with fast response time during data processing and retrieval.
- 3 Be user-friendly with a simple, clear, and responsive interface.
- 4 Ensure system reliability and availability for continuous monitoring of compliance activities.
- 5 Be scalable to support additional users, assets, and features as the system grows.

4.3 System Design

4.3.1 System Architecture

System Architecture refers to the overall structure and organization of a system, including its components, interactions, and operational processes. According to Bass (2013), system architecture defines how system elements interact to achieve the intended functionality and performance requirements. This section presents the architectural design of the proposed Facility Compliance and Safety Management System to provide a clear understanding of how the system operates and supports facility safety and regulatory compliance management. The purpose of this architecture is to ensure effective monitoring of assets, inspection management, risk assessment, incident reporting, CAPA tracking, and certification management within facilities.

The system architecture consists of the following key components:

- i. **Inspectors** conduct inspections on facility assets such as lifts, fire protection systems, electrical installations, boilers, and pressure vessels, then record inspection findings and compliance status in the system.
- ii. **Safety Officers** manage risk registers, assess identified hazards, report incidents or near-miss events, and create Corrective and Preventive Action (CAPA) tasks to address safety issues.
- iii. **Technicians** receive assigned CAPA tasks, perform corrective or preventive maintenance activities, and update task progress and completion status within the system.
- iv. **Estate Managers / Compliance Officers** manage assets, certifications, licences, inspection schedules, and assign CAPA tasks while monitoring overall compliance performance of facility assets.

v. **Administrators** manage users, system settings, and overall system operations, including monitoring reports and maintaining secure access control.

vi. The **core system** automates inspection scheduling, compliance tracking, risk monitoring, CAPA workflows, certification expiry alerts, and report generation to support continuous facility safety and regulatory compliance management.

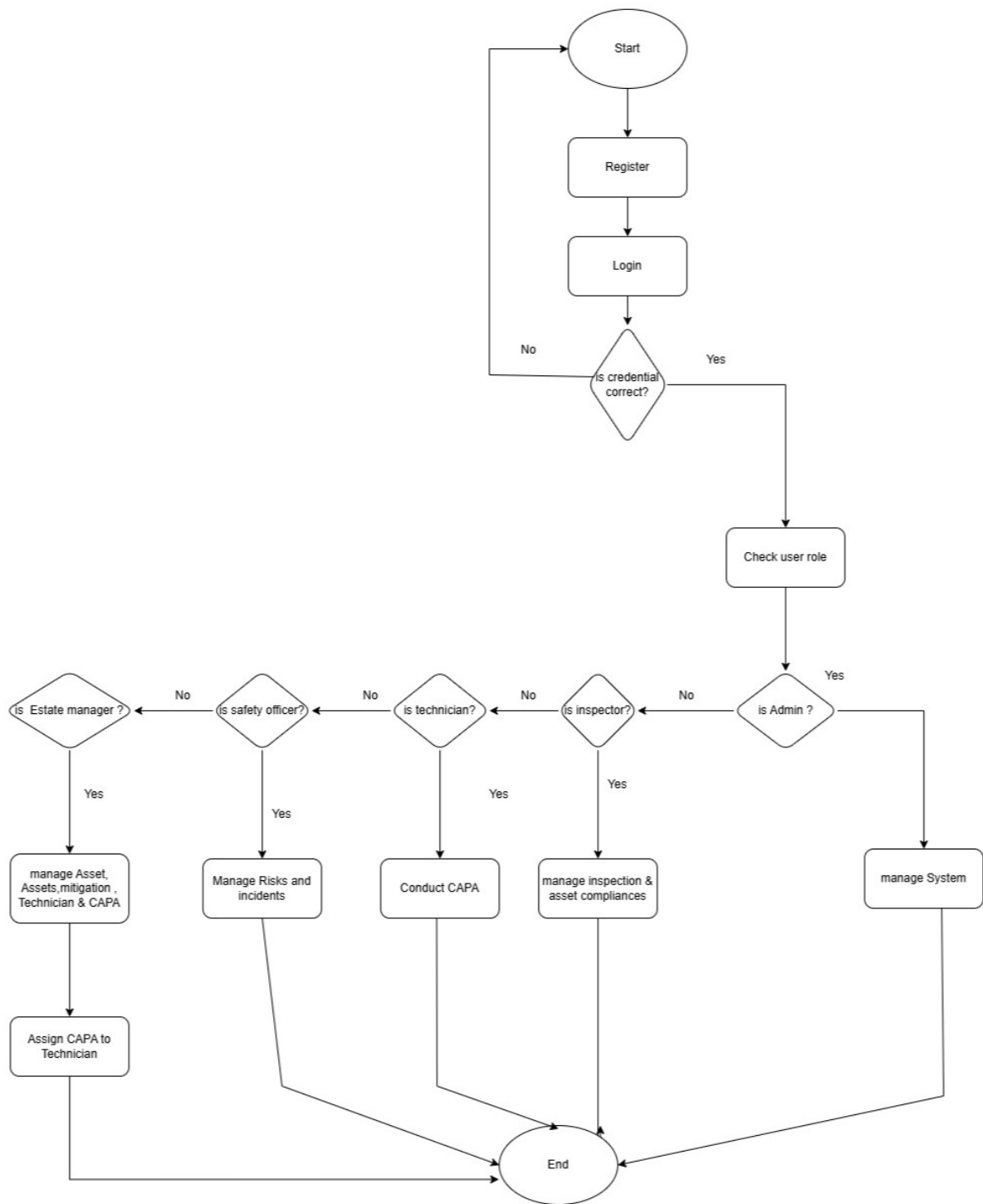
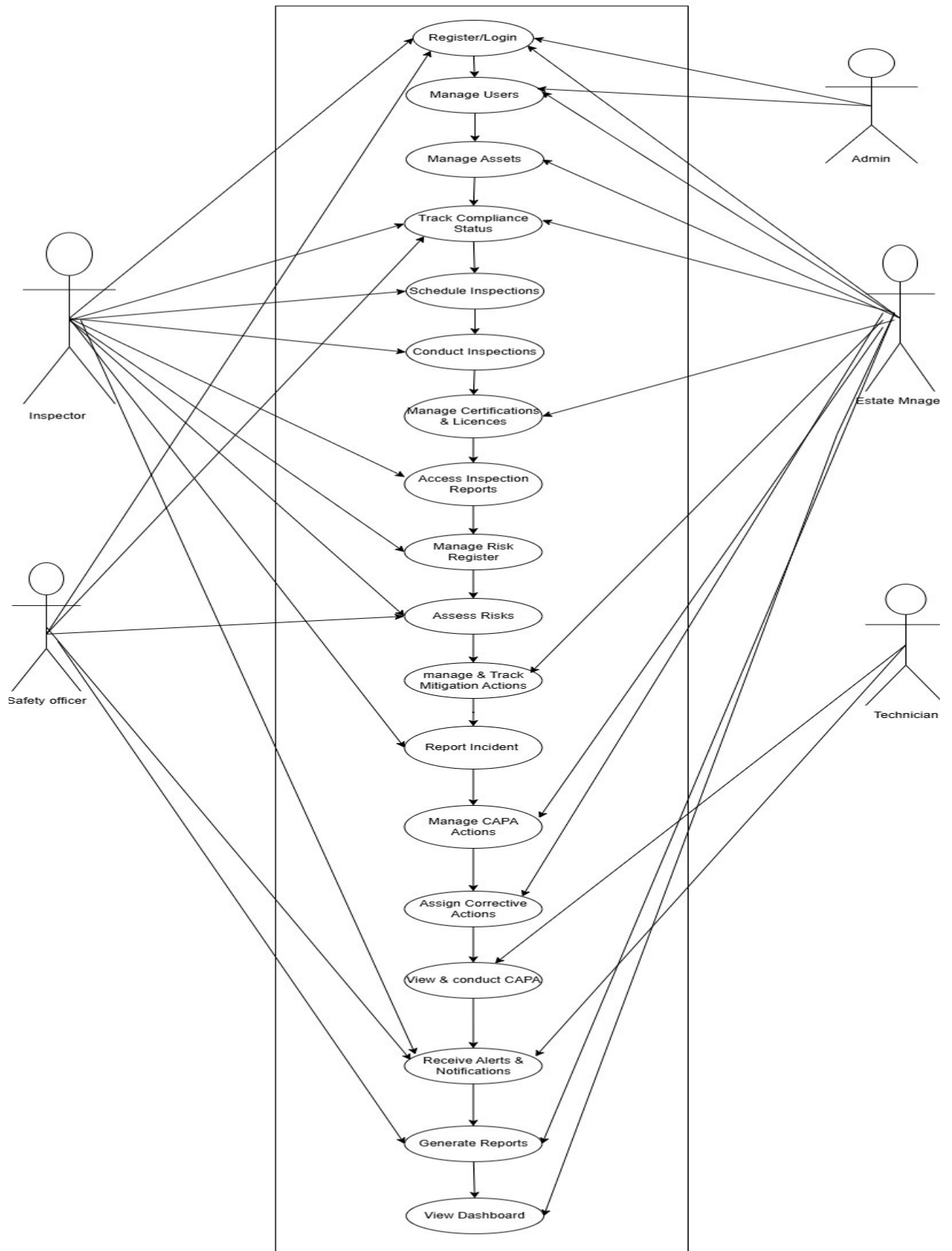


Figure 4.1: systems architecture



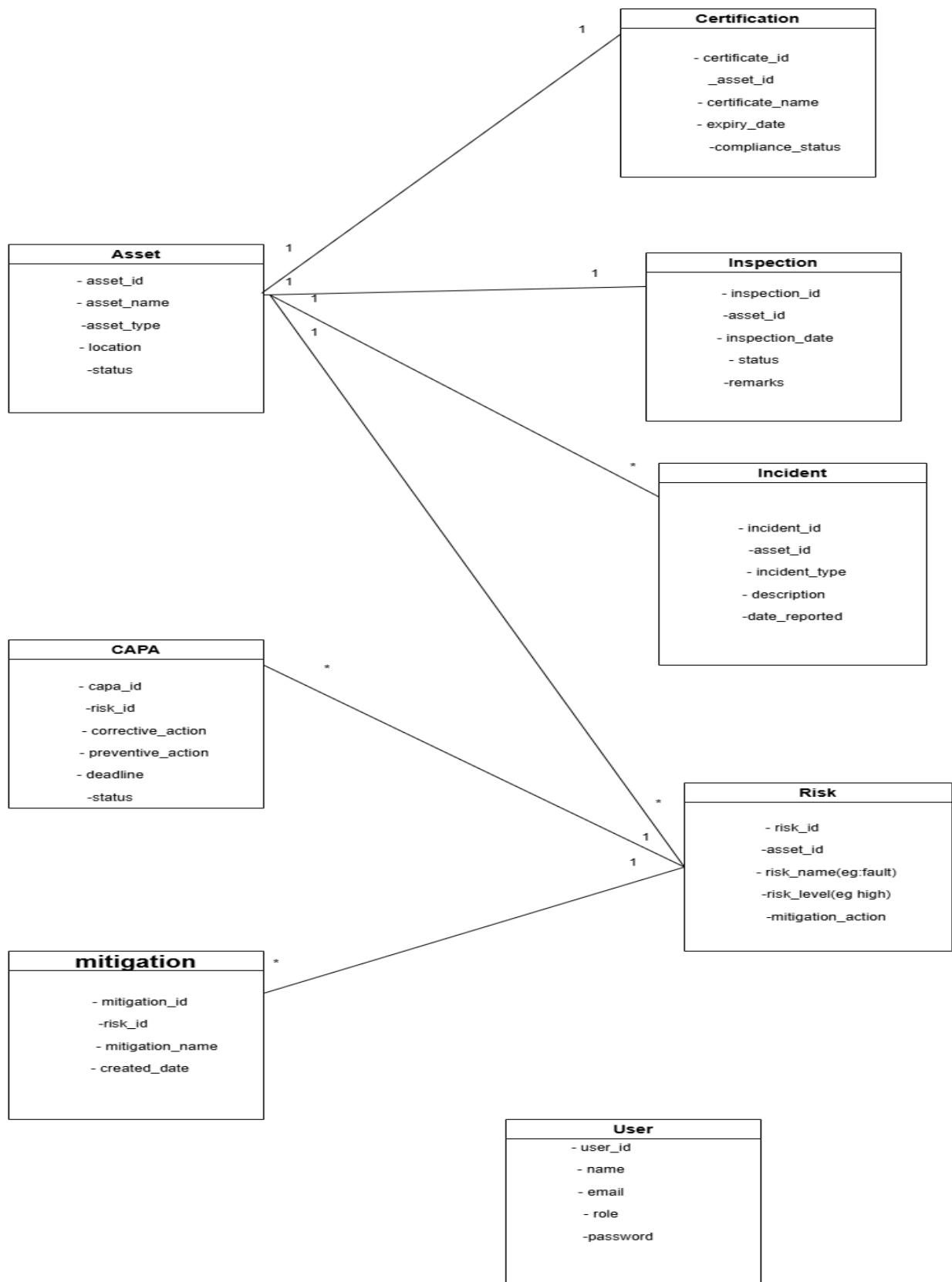


Figure 4.2:ERD diagram